PHYS 4027 / 5027 · FALL 2026 · LECTURE 19

The diffusion equation and implicit schemes

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Last time

- Von Neumann stability analysis: substitute a single wave $u_j^n = \xi^n e^{ikx_j}$ into the scheme and require $|\xi| \leq 1$ for every k.
- For FTCS on the advection equation, $|\xi|^2 = 1 + C^2 \sin^2 k\Delta x > 1$. The scheme is unstable for every time step.
- Upwind, Lax-Friedrichs and Lax-Wendroff are stable for $C = v\Delta t/\Delta x \leq 1$, the CFL condition.
- The first-order schemes smear the profile. The second-order scheme produces oscillations behind sharp features.

Part I · The diffusion equation

Where it comes from and how its solutions behave

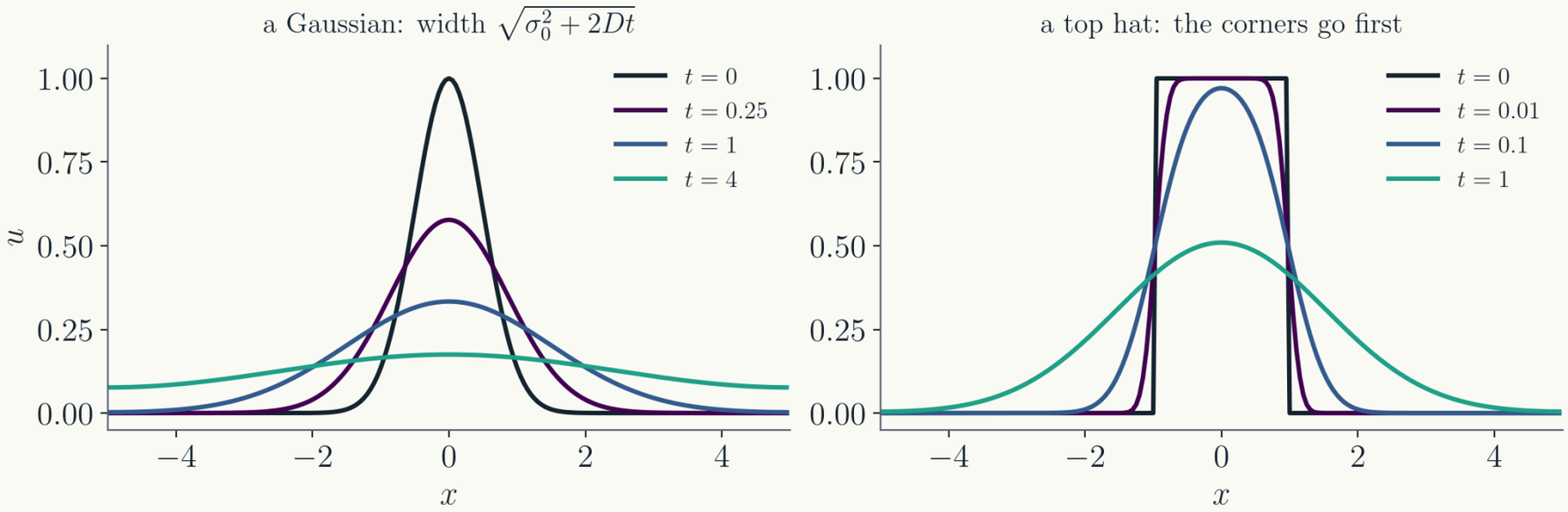
The equation

Heat conduction in a solid, dye spreading in still water, viscosity in a fluid and the decay of a magnetic field in a conductor are all described by the same equation. For a density $u(x, t)$,

$$\frac{\partial u}{\partial t} = D\,\frac{\partial^2 u}{\partial x^2}, \qquad [D] = \frac{\text{length}^2}{\text{time}}.$$

- It follows from a conservation law, $\partial_t u + \partial_x F = 0$, with a flux proportional to the gradient, $F = -D\,\partial_x u$ (Fick's law for particles, Fourier's law for heat).
- The equation is linear, first order in time and second order in space, with a single parameter D. Like the advection equation it is an initial value problem.
- The time to diffuse over a distance ℓ is ℓ^2/D. Two lengths matter in a simulation: the box, with time L^2/D, and one grid cell, with time $\Delta x^2/D$.

Behaviour of the solutions



- A single wave e^{ikx} keeps its shape and decays as e^{-Dk^2t}. Short waves decay fastest. A kink in the profile is made of short waves, so it is smoothed out immediately. The advection equation preserved kinks.
- A Gaussian remains a Gaussian, with $\sigma^2 = \sigma_0^2 + 2Dt$. The widths in the figure are $0.5, 0.87, 1.5$ and 2.9 at $t = 0$, $0.25, 1, 4$. The area under u is conserved, so the peak falls as $1/\sigma$.
- The value at each point moves toward the average of its neighbourhood, so no new maxima appear. The solution at any point depends on the entire initial profile. There are no characteristics.

Part II · FTCS on the diffusion equation

The same first scheme as for advection

The scheme

Forward difference in time, and the three-point second difference of Lecture 12 in space:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D\,\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

$$u_j^{n+1} = u_j^n + \alpha\left(u_{j+1}^n - 2u_j^n + u_{j-1}^n\right), \qquad \alpha = \frac{D\,\Delta t}{\Delta x^2}.$$

- α is the time step measured in units of the diffusion time across one cell, $\Delta x^2/D$. It plays the role that C played for advection.
- The scheme can be rearranged as $u_j^{n+1} = (1-2\alpha)\,u_j^n + \alpha\,u_{j+1}^n + \alpha\,u_{j-1}^n$, so the new value is a weighted average of the old value and its two neighbours.
- It is also Euler's method applied to the system of ODEs $du_j/dt = D\,(u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$, one equation per grid point.

Stability of FTCS

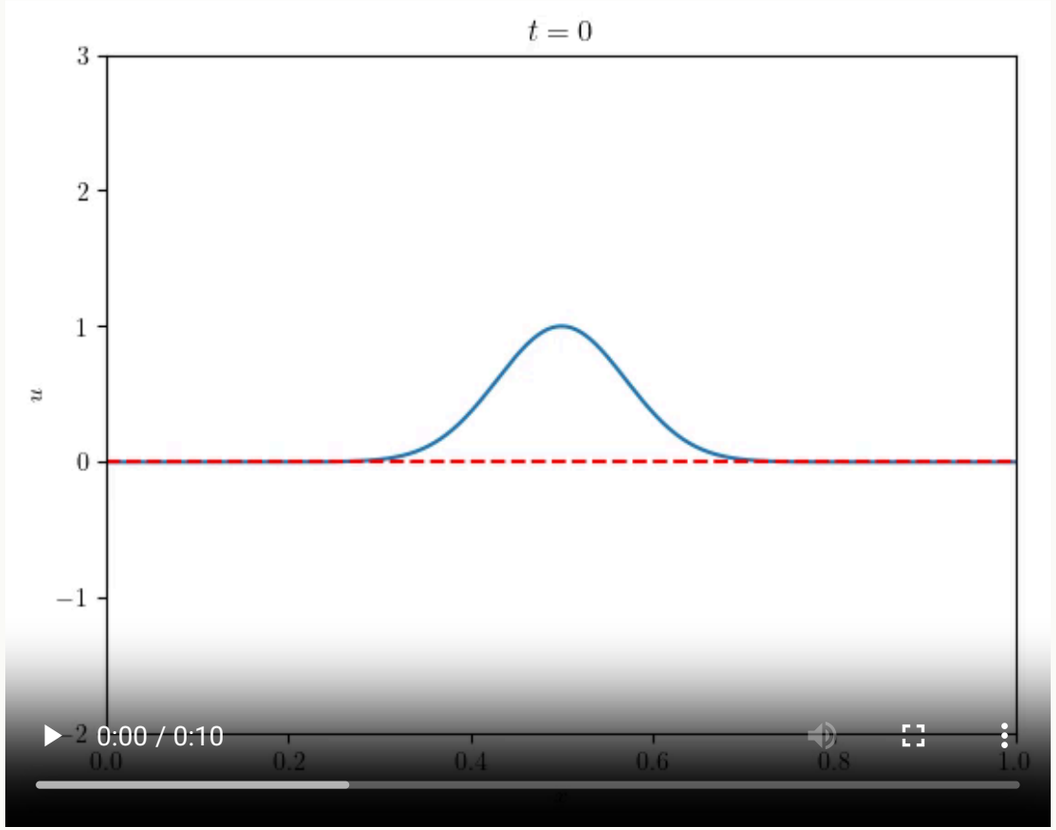
What is its stability criterion?

$$\xi = 1 - 4\alpha \sin^2\left(\frac{k\Delta x}{2}\right)$$

The scheme is stable when $\alpha \leq \frac{1}{2}$, that is $\Delta t \leq \Delta x^2 / 2D$: the time step should be less than half the diffusion time across a cell.

FTCS in action

Let's implement it: 1000 grid points on $0 \le x \le 1$, so $\Delta x = 10^{-3}$, with $D = 1$ and a Gaussian of width 0.1. Choosing $\alpha = \frac{1}{2}$, so $\Delta t = \Delta x^2 / 2D = 5 \times 10^{-7}$. The movie shows 1000 steps.



The problem with FTCS

It turns out $\Delta t \sim \Delta x^2 / D$ is too slow. If we are interested in things happening on length scales L, then we need $n \sim L^2 / \Delta x^2$ time steps! For the Gaussian of width $L = 0.1$ on the grid with $\Delta x = 10^{-3}$ that is 10^4 steps, and the movie showed 1000 of them.

How can we take much larger time steps without breaking the algorithm?

Part III · Implicit schemes

Implicit FTCS

Remember in ODEs, implicit schemes were introduced for stiff problems to overcome stability problems (Lecture 7). We can use the same concept here:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}$$

This is called the *implicit* FTCS, or BTCS, B for backward.

Implicit FTCS

What is the stability criterion for this scheme?

$$\xi = \frac{1}{1 + 4\alpha \sin^2\left(\frac{k\Delta x}{2}\right)}$$

$|\xi| < 1$ for any Δt. The implicit FTCS scheme is *unconditionally stable*.

Implicit FTCS

However, there is only one problem:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = D\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}$$

This is a system of linear equations with u_j^{n+1} being the unknown vector. Can we write down the matrix A?

Implicit FTCS

The matrix A associated with the implicit FTCS scheme is:

$$A = \begin{pmatrix} 1+2\alpha & -\alpha & 0 & \cdots & 0 & 0 & 0 \\ -\alpha & 1+2\alpha & -\alpha & \cdots & 0 & 0 & 0 \\ 0 & -\alpha & 1+2\alpha & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1+2\alpha & -\alpha & 0 \\ 0 & 0 & 0 & \cdots & -\alpha & 1+2\alpha & -\alpha \\ 0 & 0 & 0 & \cdots & 0 & -\alpha & 1+2\alpha \end{pmatrix}$$

This is a tri-diagonal matrix that is diagonally dominant.

Implicit FTCS

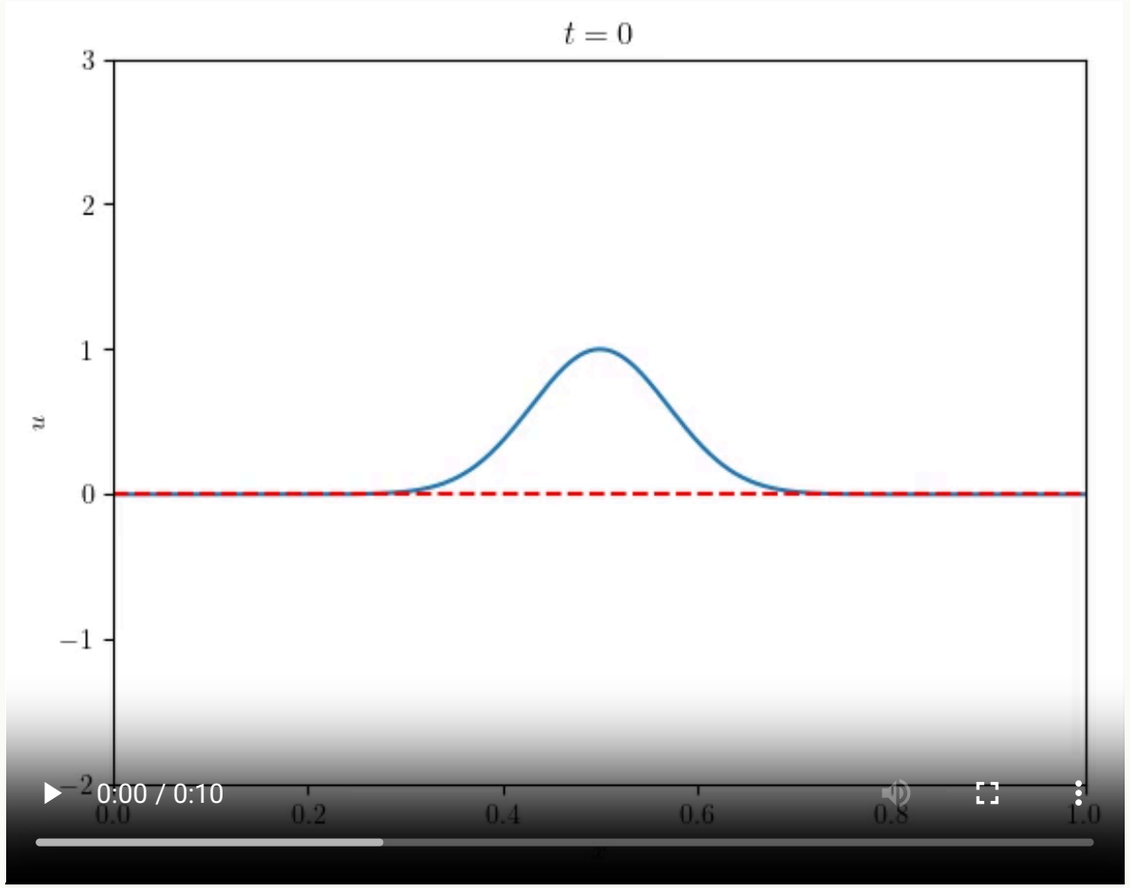
At every time step, we solve the linear system:

$$\mathbf{A}\mathbf{u}^{n+1} = \mathbf{u}^n$$

Since $\mathbf{A}$ is tri-diagonal, we can use the Thomas algorithm (Lecture 14), the solver you wrote in Homework 7. $\mathbf{A}$ never changes over time, so we find its LU decomposition once and reuse it at every time step. One step costs $O(N)$, about twice an FTCS step.

Implicit FTCS in action

Let's see how it does. Same grid and initial profile, choosing $\alpha = 100$, so $\Delta t = 10^{-4}$, two hundred times the FTCS limit. The movie shows 100 steps, to $t = 10^{-2}$.



Part IV · Crank-Nicolson

Second order accuracy in time

The implicit FTCS scheme is still 1st order accurate in time. To achieve 2nd order accuracy in time, we can take the average between the FTCS and implicit FTCS schemes:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{D}{2}\left(\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} + \frac{u_{j+1}^{n} - 2u_j^{n} + u_{j-1}^{n}}{\Delta x^2}\right)$$

Why is this scheme second order accurate in time?

This numerical scheme is called the *Crank-Nicolson* scheme, and is the recommended scheme for diffusion-type PDEs.

Crank-Nicolson scheme

Let's work out the amplification factor:

$$\xi = \frac{1 - 2\alpha \sin^2\left(\frac{k\Delta x}{2}\right)}{1 + 2\alpha \sin^2\left(\frac{k\Delta x}{2}\right)}$$

Again, this scheme is unconditionally stable.

Crank-Nicolson scheme

Again, one needs to solve a tri-diagonal system at every time step. However, this time the right hand side is not simply $\mathbf{u}^n$:

$$\mathbf{A}\mathbf{u}^{n+1} = \mathbf{B}\mathbf{u}^n$$

The matrix $\mathbf{A}$ has $\left(-\frac{\alpha}{2}, 1+\alpha, -\frac{\alpha}{2}\right)$ on its diagonals, while the matrix $\mathbf{B}$ has $\left(\frac{\alpha}{2}, 1-\alpha, \frac{\alpha}{2}\right)$ on its diagonals, with the same $\alpha = D\Delta t/\Delta x^2$.

Part V · The Schrödinger equation

The Schrödinger equation

The Schrödinger equation is the fundamental equation of quantum mechanics:

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi$$

It is like a diffusion equation in imaginary time, so the methods today are applicable.

The Schrödinger equation

We first get rid of the units by performing non-dimensionalization, the equation looks like:

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2}\frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$$

Let's use the implicit FTCS scheme:

$$i\frac{\psi_j^{n+1} - \psi_j^n}{\Delta t} = -\frac{1}{2}\frac{\psi_{j+1}^{n+1} - 2\psi_j^{n+1} + \psi_{j-1}^{n+1}}{\Delta x^2} + V(x)\psi_j^{n+1}$$

The Schrödinger equation

The amplification factor for the implicit FTCS scheme is:

$$\xi = \frac{1}{1 + i\left[\frac{2\Delta t}{\Delta x^2}\sin^2\left(\frac{k\Delta x}{2}\right) + V_j \Delta t\right]}$$

The scheme is unconditionally stable. However, $|\xi| < 1$ in general, so it does not preserve amplitudes. In quantum mechanical language, it is not *unitary*.

The Schrödinger equation

Unitarity is important in quantum mechanics. The wave function is understood as a probability amplitude. The probability of finding the particle at location x is $P(x) = |\psi(x)|^2$. It satisfies the normalization condition:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

Amplitude decay is not a desirable trait of the numerical method when solving quantum mechanics!

The Schrödinger equation

If we formally write the Schrödinger equation as:

$$i\frac{\partial \psi}{\partial t} = \hat{H}\psi$$

where $\hat{H}$ is the Hamiltonian operator, then the implicit FTCS scheme is effectively:

$$\psi^{n+1} = \left(1 + i\hat{H}\Delta t\right)^{-1}\psi^{n}$$

$(1 + i\hat{H}\Delta t)^{-1}$ is a non-unitary approximation to the time evolution operator $e^{-i\hat{H}\Delta t}$.

The Schrödinger equation

The cure is to use the so-called "Cayley's form" for $e^{-i\hat{H}\Delta t}$:

$$e^{-i\hat{H}\Delta t} \approx \frac{1 - \frac{1}{2}i\hat{H}\Delta t}{1 + \frac{1}{2}i\hat{H}\Delta t}$$

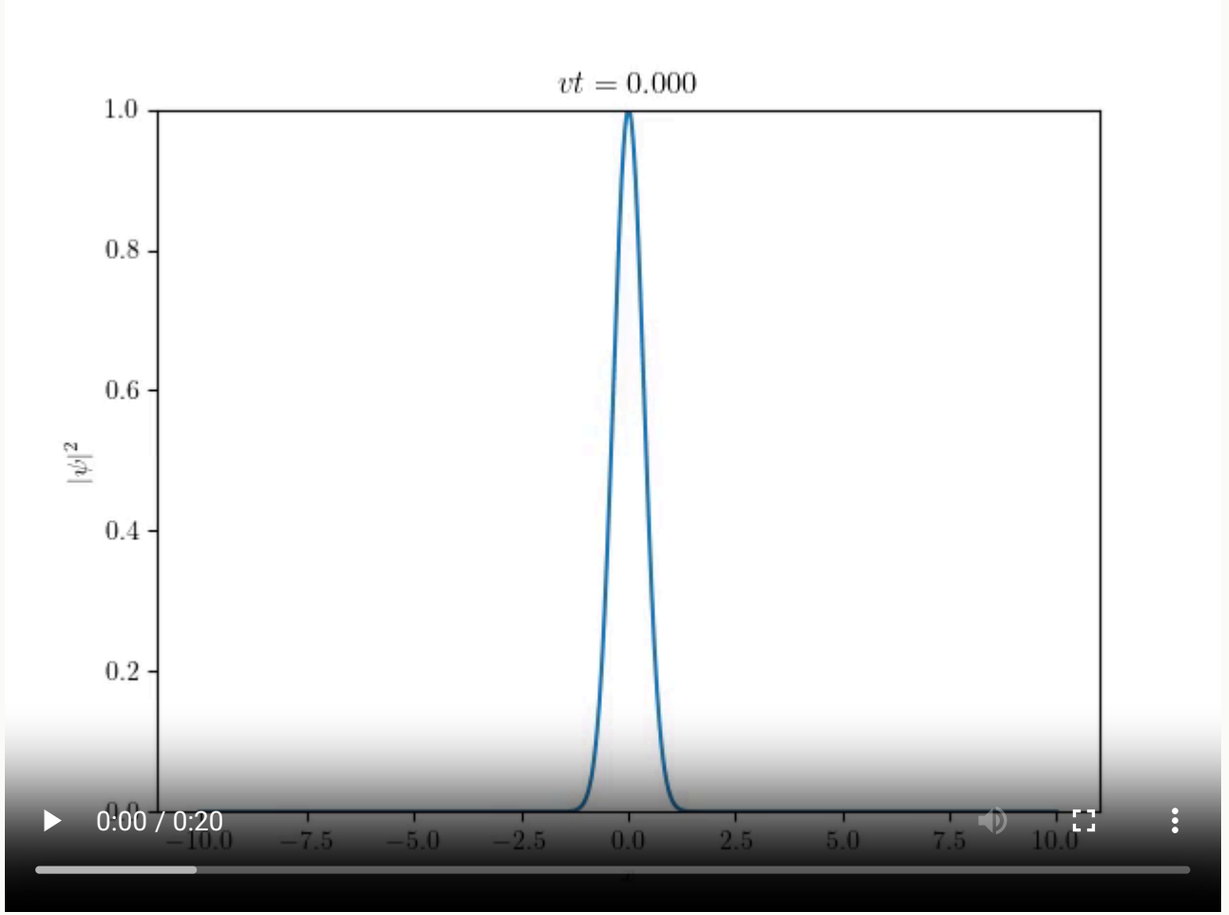
Then the numerical scheme becomes:

$$\left(1 + \frac{1}{2}i\hat{H}\Delta t\right)\psi_j^{n+1} = \left(1 - \frac{1}{2}i\hat{H}\Delta t\right)\psi_j^n$$

In fact, this is equivalent to the Crank-Nicolson scheme!

The Schrödinger equation

Let's see the Crank-Nicolson scheme in action:



Summary

- FTCS is stable for the diffusion equation when $\Delta t \leq \Delta x^2 / 2D$, which is too small a step.
- Implicit FTCS (BTCS) is unconditionally stable. Each step is a tri-diagonal solve.
- Crank-Nicolson averages the two, is unconditionally stable and second order in time.
- For the Schrödinger equation Crank-Nicolson is also unitary.

Next time: the wave equation on a staggered grid, and then Poisson's equation.